

WON KO

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SUMMARY

Senior Researcher in autonomous robotics specializing in LiDAR–visual–inertial localization, state estimation, and robust multi-sensor localization. Experience includes LIO-SAM / SC-LIO-SAM, LVI-SAM evaluation / LIO–VIO integration, GTSAM/iSAM2, and vision- and intensity-assisted localization on real AMR and autonomous-forklift platforms.

TECHNICAL SKILLS

Programming / Data	C/C++, Python, OpenCV; localization/trajectory analysis, timing, RMSE, and repeatability evaluation
Localization / SLAM	LIO-SAM, SC-LIO-SAM, LVI-SAM evaluation / LIO–VIO integration, LiDAR–visual–inertial localization, relocalization, GICP, Scan Context
Visual / Intensity Localization	RGB-D / visual feature integration, depth-supported 3D features, Intensity Scan Context, calibrated LiDAR-intensity priormaps, vision- and intensity-assisted localization
State Estimation / Optimization	GTSAM/iSAM2, IMU preintegration, ESKF/EKF/Kalman filtering, nonlinear least-squares, LOAM scan-to-map optimization, degeneracy handling
Calibration / Sensor Integration	IMU/inertial-sensor calibration (accel/mag bias–scale, ellipsoid fitting, axis misalignment, gyro-bias estimation); single-/two-pose monocular camera calibration; PnP/RANSAC 2D–3D pose estimation; LiDAR/base SE(2), synchronization and TF integration
Robotics Systems	ROS / ROS 2, TF2, PCL, Linux, Git, Docker, colcon, CI/CD; real-robot deployment and validation

EDUCATION

University of California, Santa Cruz Master of Science in Computer Engineering — Robotics and Control M.S. Project: “Autonomously Driving Tricycle with Monocular Vision” Advisor: Prof. Dejan Milutinović	August 2022
University of California, Santa Cruz Bachelor of Science in Computer Engineering — Robotics and Control	December 2019

TEACHING EXPERIENCE

Teaching Assistant / University of California, Santa Cruz <i>Mechatronics (ECE 118)</i>	Spring 2022, Fall 2021 <i>Santa Cruz, CA</i>
• Led lab sections on event-driven programming, hierarchical state machines (HSMs), sensor interfacing, and end-to-end system integration; provided debugging support and graded coursework with the TA team. • Maintained and repaired lab hardware; performed safety checks and managed spare parts/inventory to minimize downtime.	
Teaching Assistant / University of California, Santa Cruz <i>Computer Architecture (CSE 120)</i>	Spring 2021, Winter 2022 <i>Santa Cruz, CA</i>
• Led RISC-V and hardware design labs using Verilog/SystemVerilog; guided students through simulation, debugging, and design verification. • Held weekly office hours and graded assignments/exams; supported performance analysis and correctness debugging for lab implementations.	

RESEARCH EXPERIENCE

Senior Researcher / Future Tech R&D Lab, Mobyus

November 2024 - Present

- Research, develop, and validate multi-sensor localization and autonomous robotics systems for AMR and autonomous-forklift platforms on real warehouse hardware, spanning mapping, perception, planning, and control.
- Improved reliability of a LIO-SAM / SC-LIO-SAM-derived ROS 2 stack with GTSAM/iSAM2, IMU-preintegration recovery, scan-to-map safeguards, abnormal-motion diagnostics, and map-integrity protection.
- Designed and implemented intensity-assisted prior-map localization using calibrated LiDAR intensity, keyframe-derived priormaps, Intensity Scan Context prescoring, and small-GICP geometric validation.
- Extended the LIO backbone with synchronized RGB-D/visual feature projection and keyframe persistence to evaluate camera-assisted weighting, mapping, and future relocalization without replacing the LiDAR-inertial pose backbone.

Researcher / Robotics and Control Lab, UC Santa Cruz

September 2020 - August 2023

- Researched probabilistic state estimation and stochastic nonlinear control for GPS-free navigation under sensor noise and model uncertainty, including stochastic optimal Dynamic Programming.
- Designed EKF and second-order nonlinear state estimators for mobile-robot pose, velocity, and heading from noisy camera position measurements; modeled process/measurement covariance and evaluated noise sensitivity in MATLAB.
- Performed system identification of an unknown propeller-actuated mechanism in V-REP/MATLAB, deriving discrete/continuous transfer functions and estimating servo/beam dynamics with batch least squares and recursive least squares.
- Conducted MPU-9250 inertial-sensor calibration and attitude-estimation experiments using accel/mag bias-scale correction, 3D ellipsoid tumble calibration, Wahba/SVD axis alignment, and DCM-based gyroscope-bias correction.

SELECTED LOCALIZATION & SLAM R&D

Robust Prior-Map Localization with Calibrated LiDAR Intensity

Ongoing R&D

- Built incident-angle/range-aware reflectivity calibration and keyframe-derived intensity priormaps for geometry-consistent global localization in repetitive warehouse areas.
- Implemented 20×60 Intensity Scan Context with bounded candidate/yaw prescoring; calibrated intensity ranks hypotheses before small-GICP geometric acceptance.
- Instrumented intensity MAE/RMSE and localization diagnostics; publication-oriented Recall@K, false-match, pose-error, and perturbation testing is in progress.

Camera-LiDAR Integration and Visual-Aided LIO

Ongoing Prototype

- Defined camera-LiDAR mounting/TF integration with LIO-SAM kept as the authoritative robot pose, avoiding competing VIO TF trees while attaching visual outputs to the camera frame.
- Added synchronized RGB-D/depth-supported visual feature projection into the LIO pipeline with timing, FOV/depth association, confidence, conflict, and occlusion diagnostics for safe visual weighting.
- Persisted per-keyframe visual depth/RGB and corner/surface/pole clouds and rebuilt global visual maps; asynchronous RGB updates preserve depth/feature processing when color arrives later.

Autonomous Forklift System Development & Validation

November 2024 - October 2025

- Implemented camera-to-ground projective calibration and planar LiDAR/base SE(2) trajectory calibration with before/after position/yaw RMSE reporting and report-only configuration previews.
- Built and validated an ESKF-based LiDAR-IMU localization stack on autonomous forklifts, achieving ≤ 30 mm position error in controlled testbed evaluation; integrated navigation, Frenet/B-spline trajectory tracking, pallet keypoint perception, and depth-based free-space estimation.